

Experiment No. – 3

Aim –

- (A.) You are given with the Employee schema; the task is to find the highest score for each respective department. For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28. Same with the other departments i.e., D2, D3 and D4.
- (B.) A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.
- (C.) A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

Tools Required

- Postgres
- PgAdmin4

Queries:

A.

```
CREATE TABLE Employee(  
    EID INT PRIMARY KEY,  
    DEPT VARCHAR(10),  
    SCORES DECIMAL(5,2));  
  
INSERT INTO Employee (EID, DEPT, SCORES)  
VALUES  
(1, 'D1', 1.00),  
(2, 'D1', 5.28),  
(3, 'D1', 4.00),  
(4, 'D2', 8.00),
```

```
(5, 'D1', 2.50),  
(6, 'D2', 7.00),  
(7, 'D3', 9.00),  
(8, 'D4', 10.20);
```

```
UPDATE Employee as e  
set SCORES = (  
SELECT MAX(SCORES)  
FROM Employee  
where DEPT = e.DEPT  
);
```

B.

```
CREATE TABLE SalesPerson  
(  
    seller_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    city VARCHAR(50),  
    commission INT  
);  
  
CREATE TABLE Company(  
    com_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    city VARCHAR(50)  
);  
  
CREATE TABLE Orders(  
    order_id INT PRIMARY KEY,  
    order_date DATE,  
    com_id INT,  
    seller_id INT,  
    amount INT,  
    FOREIGN KEY (com_id)  
        REFERENCES Company(com_id),
```

```
FOREIGN KEY (seller_id)
REFERENCES SalesPerson(seller_id)
);
```

```
INSERT INTO SalesPerson
VALUES
```

```
(1,'John','New York',15),
(2,'Amy','Los Angeles',13),
(3,'Mark','Chicago',12),
(4,'Pam','Boston',15);
```

```
INSERT INTO Company
VALUES
```

```
(1,'RED','Boston'),
(2,'ORANGE','New York'),
(3,'YELLOW','Boston'),
(4,'GREEN','Austin');
```

```
INSERT INTO Orders
VALUES
```

```
(1,'2024-01-10',1,1,1200),
(2,'2024-01-12',2,1,800),
(3,'2024-01-15',3,2,2500),
(4,'2024-01-18',1,3,1500),
(5,'2024-01-22',4,2,700),
(6,'2024-01-25',2,3,2000),
(7,'2024-01-28',3,4,3000),
(8,'2024-01-30',4,4,200);
```

```
select seller_id from orders
group by seller_id
```

```
having sum(amount)=
```

```
(select max(total) from (
```

```
SELECT seller_id ,sum(amount) as total from Orders
Group by seller_id ));
```

C.

```
CREATE TABLE Department(  
    id INT PRIMARY KEY,  
    name VARCHAR(50)  
);
```

```
INSERT INTO Department (id, name)  
VALUES  
(1, 'IT'),  
(2, 'Sales');
```

```
CREATE TABLE Employee(  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    salary INT,  
    departmentId INT,  
    FOREIGN KEY (departmentId)  
    REFERENCES Department(id)  
);
```

```
INSERT INTO Employee (id, name, salary, departmentId)  
VALUES  
(1, 'Joe', 85000, 1),  
(2, 'Henry', 80000, 2),  
(3, 'Sam', 60000, 2),  
(4, 'Max', 90000, 1),  
(5, 'Janet', 69000, 1),  
(6, 'Randy', 85000, 1),  
(7, 'Will', 70000, 1);
```

```
select  
d.name as DEPARTMENT ,  
e.name as EMPLOYEE ,
```

```

e.salary as SALARY
from Employee as e
Join Department as d
ON e.departmentId = d.id
where (
select count(distinct e2.salary)
from employee as e2
where e2.departmentId = e.departmentId
and e2.salary > e.salary ) < 3 ;

```

Outputs:

	eid [PK] integer	dept character varying (10)	scores numeric (5,2)
1	1	D1	5.28
2	2	D1	5.28
3	3	D1	5.28
4	4	D2	8.00
5	5	D1	5.28
6	6	D2	8.00
7	7	D3	9.00
8	8	D4	10.20

	seller_id integer
1	3

	department character varying (50)	employee character varying (50)	salary integer
1	IT	Joe	85000
2	Sales	Henry	80000
3	Sales	Sam	60000
4	IT	Max	90000
5	IT	Randy	85000
6	IT	Will	70000

Learning Outcome

Create and populate database tables using SQL.

Understand primary keys and data types.

Retrieve records using SELECT.

Perform data analysis using aggregate functions (AVG, SUM, MAX, MIN, etc.) and department-wise operations.

Perform table joins to combine related data.

Use correlated subqueries for advanced filtering